

Code2Day + MentorMind: Cost-Minimised Deployment

Combined monthly cost for **both products on one shared AWS estate**, at **3,000 peak concurrent users**, modelled against an **8-hour daily usage window**. Two population sizes; every reasonable cost lever applied.

TIER A · 3,000 REGISTERED ₹76,500/mo \$900/month · 3,000 concurrent · 3,000 DAU Every student online at peak — 100% concurrency ₹25.50 per student/month	TIER B · 10,000 REGISTERED ₹1,46,965/mo \$1,729/month · 3,000 concurrent · ~7,000 DAU 30% concurrency — the normal campus pattern ₹14.70 per student/month
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The headline finding: Tier B costs 1.9× more in total but 42% less per student

Both tiers need **identical peak capacity** — 3,000 concurrent users is 3,000 concurrent users regardless of how many are on the roll. The expensive part of this platform is that peak, and Tier B spreads it across 3.3× more students. Tier A pays **₹25.50/student** for 100% concurrency; Tier B pays **₹14.70**.

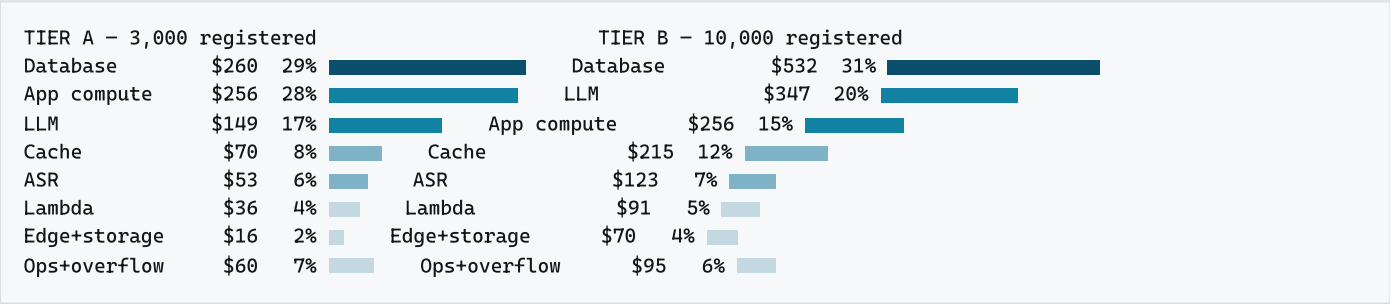
Practical consequence: **if the roll is going to grow to 10,000, build for it now**. The infrastructure in §4 serves both tiers; only the database class, storage and AI volume differ. Growing from Tier A to Tier B adds ₹70,465/month of cost but 7,000 students.

REPORT DATE	9 September 2026
SCOPE	Combined: Code2Day and MentorMind on one estate. Per-product attribution in §6
REGION	ap-south-1 (Mumbai)
FX BASIS	₹85.00 / USD
USAGE WINDOW	8 hours/day × 22 weekdays = 176 peak-hours/month (24% of the month). This assumption drives the largest single saving in the report — see §3
OBJECTIVE	Minimise cost while still serving 3,000 concurrent users without queueing. §5 lists what was traded away to get there
PRICE BASIS	m7g.2xlarge at \$0.2333/hr ap-south-1 VERIFIED RDS, ElastiCache and ALB rates are baseline published rates with RI discounts applied VERIFY IN CALCULATOR AI rates are September 2026 list prices VERIFIED
COMPANION TO	AWS_DEPLOYMENT_OPTIONS.md , PROJECT_DEPLOYMENT_PROFILE.md , MENTORMIND_MODULE_COST_BREAKDOWN.md

1 Summary

TABLE 1 — COMBINED MONTHLY COST, BOTH TIERS

Cost centre	Tier A	share	Tier B	share
Database tier (RDS + replicas + Proxy + storage)	\$260	29%	\$532	31%
Application compute (EC2 + ALB)	\$256	28%	\$256	15%
LLM inference (DeepSeek V3.2 via Mantle)	\$149	17%	\$347	20%
Cache tier (ElastiCache for Valkey)	\$70	8%	\$215	12%
Speech-to-text (Groq Whisper turbo)	\$53	6%	\$123	7%
Serverless execution (Lambda + SQS)	\$36	4%	\$91	5%
Edge + storage (CloudFront, S3)	\$16	2%	\$70	4%
Operations (CloudWatch, DNS, backup, IPv4)	\$45	5%	\$65	4%
AI overflow allowance	\$15	2%	\$30	2%
Total per month (USD)	\$900	100%	\$1,729	100%
Total per month (INR)	₹76,500		₹1,46,965	
Per registered student per month	₹25.50		₹14.70	



1.1 Fixed versus variable, per student

The split matters for planning, because only one half moves when the roll changes.

Component class	Tier A total	₹/student	Tier B total	₹/student
Capacity-driven — sized by the 3,000 peak, not the roll (EC2, ALB, cache, ops)	\$371	₹10.51	\$536	₹4.56
Population-driven — scales with students and usage (database size, storage, AI, Lambda)	\$529	₹14.99	\$1,193	₹10.14
Total	\$900	₹25.50	\$1,729	₹14.70

Capacity-driven cost per student falls 57% between the tiers (₹10.51 → ₹4.56) because the same peak serves more people. Population-driven cost per student falls only 32%, and would be flat were it not for the smaller database class Tier A can use.

2 Load model

These are stated assumptions, not measurements

PROJECT_DEPLOYMENT_PROFILE.md:1530 lists DAU and peak concurrency as "UNKNOWN — requires measurement."
Everything below scales linearly off the assumptions in Table 2. §8.1 gives the three commands that replace them with facts.
Run those before committing budget.

TABLE 2 — LOAD PARAMETERS

Parameter	Basis	Tier A	Tier B
Registered students	stated requirement	3,000	10,000
Peak concurrent users	stated requirement	3,000	3,000
Implied concurrency ratio	concurrent ÷ registered	100%	30%
Daily active users	Tier A: all students, by definition. Tier B: 70% of roll	3,000	7,000
Daily usage window	stated: "8 hours or something"	8 h	8 h
Peak-hours per month	8 h × 22 weekdays	176 h	176 h
Off-peak hours per month	730 – 176	554 h	554 h
Peak API request rate	0.2 req/s per concurrent user	600/s	600/s
Design target	×1.5 headroom for burst and node loss	900/s	900/s
Graded voice exercises	3 × ~60 s per active user/day	3 min	3 min
Voice adoption	40% of DAU record audio on a given day	40%	40%
Monthly audio	DAU × 40% × 3 min × 22	79,200 min 1,320 hr	184,800 min 3,080 hr
Monthly LLM calls	after the §7.2 consolidation	~114,000	~267,000

2.1 What Tier A actually describes

3,000 registered *and* 3,000 concurrent means every student on the roll is online at the same moment. That is not a browsing pattern; it is a **synchronised event** — a scheduled lab, a placement test, a timed contest. It is the most demanding shape a platform of this size can have, and it is why Tier A's per-student cost is high despite being the smaller deployment. Tier B's 30% ratio is the normal campus pattern, where usage staggers across the day.

Both tiers are sized for the same 3,000-concurrent peak, so **Tier A is not a cheaper architecture — it is the same architecture with fewer students paying for it.**

3 The 8-hour window is the biggest lever

Peak capacity is needed for **176 of 730 hours a month — 24%**. Provisioning for peak around the clock wastes three-quarters of the compute spend. Scheduled scaling on the Auto Scaling group, combined with a Savings Plan sized to the *baseline* rather than the peak, is where the single largest saving in this report comes from.

EC2 application fleet – m7g.2xlarge at \$0.2333/hr (ap-south-1, verified)

(a) Naive: 4 nodes on-demand, 24/7
 $4 \times 730 \text{ h} \times \$0.2333 = \$681/\text{mo}$

(b) Savings Plan on all 4 nodes, 24/7 (3-yr, ~62% off)
 $4 \times 730 \text{ h} \times \$0.2333 \times 0.38 = \$259/\text{mo}$

(c) CHOSEN: 2 nodes baseline on Savings Plan + 2 burst on-demand for 176 h
baseline $2 \times 730 \text{ h} \times \$0.2333 \times 0.38 = \$129$
burst $2 \times 176 \text{ h} \times \$0.2333 = \$82$
 $= \$212/\text{mo}$

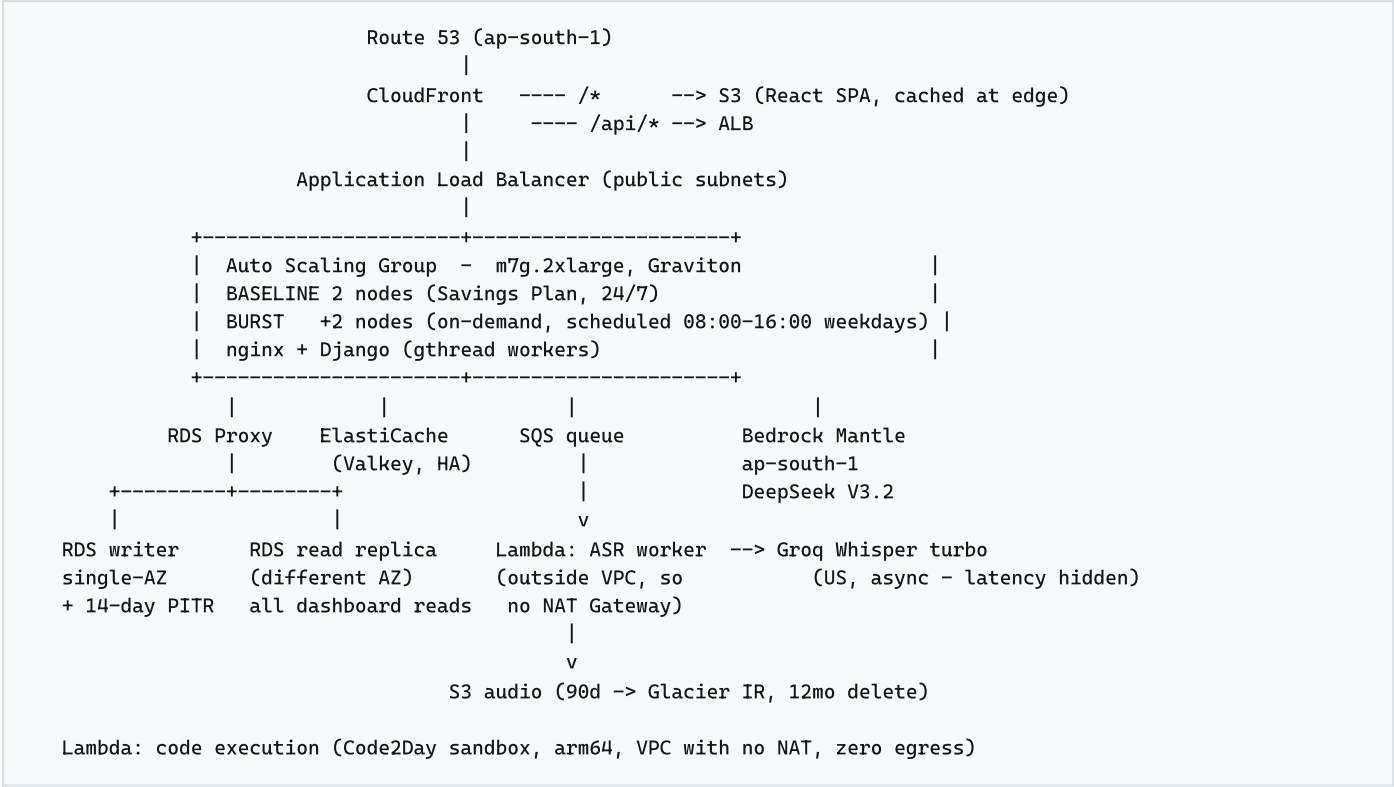
Saving of (c) over (a): \$469/mo – 69%. Over (b): \$47/mo – 18%.

Option (c) is chosen because it keeps **two nodes running at all times** – so the platform stays available and redundant outside the window – while paying on-demand rates only for the burst capacity that exists for 24% of the month. Committing a Savings Plan to all four nodes (option b) would waste the commitment for 554 hours a month.

Where the 8-hour window does *not* help

The database and cache cannot follow it. RDS Multi-AZ cannot be stopped, and a stopped single-AZ instance restarts automatically after 7 days. ElastiCache nodes are likewise always-on. Together that is \$330–\$747/month of unavoidable 24/7 spend – 37% of Tier A and 43% of Tier B. §4.3 covers the one alternative (Aurora Serverless v2) and why it was not chosen.

4 Architecture



4.1 Sizing rationale

Decision	Reasoning
m7g.2xlarge × 4 at peak	900 req/s ÷ 40 req/s per vCPU = 23 vCPU. Four 8-vCPU nodes give 32 vCPU — 28% headroom plus N+1 node redundancy. 40 req/s/vCPU assumes Redis-cached dashboards and replica-served reads (\$7.3)
Graviton throughput	~20% better price/performance, and ap-south-1 rates are lower than us-east-1 for m7g . Requires dropping sentence-transformers — see §7.3 #4
Single-AZ writer + PITR	Multi-AZ doubles the writer cost for failover in ~1 minute instead of ~30. For a campus platform outside placement season, PITR plus a promotable replica is a defensible trade. Upgrade cost in \$5.2
Read replica in a second AZ	Serves all dashboard, leaderboard and analytics reads, and doubles as a promotion target if the writer's AZ fails
RDS Proxy	Not optional. Code2Day sets no CONN_MAX_AGE (PROJECT_DEPLOYMENT_PROFILE.md :215), so Django opens a fresh connection per request. At 900 req/s that alone exhausts the database
ASR Lambda outside the VPC	It needs only S3 and the Groq API, both public. Keeping it out of the VPC means no NAT Gateway — a clean \$33–41/month saving
App nodes in public subnets	Public IPv4 at \$3.60/node-month is far cheaper than NAT or a set of interface endpoints. Security groups admit ALB traffic only; IMDSv2 with hop limit 1
Valkey, not Redis OSS	Same functionality, materially lower node and serverless rates

4.2 Why not Aurora Serverless v2

An 8-hour duty cycle looks like the textbook case for Aurora Serverless v2 at \$0.12/ACU-hour with scale-to-zero, and it was costed. It loses on two counts.

Option	Tier A	Tier B	Note
RDS provisioned + 1-yr RI CHOSEN	\$260	\$532	Predictable, no I/O exposure
Aurora Serverless v2, Standard storage	\$420 + I/O	\$706 + I/O	The I/O trap. At \$0.20 per million I/Os, 900 req/s could add \$500–2,000/month
Aurora Serverless v2, I/O-Optimized	\$551	\$938	Removes the I/O charge for +30% on ACU — still 2× and 1.8× the chosen option

Serverless v2 is cheaper than provisioned when utilisation sits below roughly 40% of peak — which this workload's *hours* do. But ACU pricing carries no reserved-instance discount, so it is competing against an RI-discounted provisioned instance, and I/O-Optimized adds 30% on top. **Revisit only if a measured off-peak load turns out to be genuinely near-idle and the peak ACU requirement proves lower than the 8–16 ACU assumed here.**

5 Detailed cost breakdown

TABLE 3 — COMPONENT-BY-COMPONENT MONTHLY COST, BOTH TIERS

Component	Configuration	Rate basis	Tier A USD	Tier B USD
APPLICATION COMPUTE				
EC2 baseline	2 × m7g.2xlarge, 24/7	\$0.2333/hr, 3-yr Compute Savings Plan (~62% off)	129	129
EC2 burst	+2 × m7g.2xlarge, 176 h/mo	\$0.2333/hr on-demand, scheduled scaling	83	83
Application Load Balancer	1 ALB, ~16 LCU at peak	\$0.0225/hr + \$0.008/LCU-hr	44	44
Compute subtotal			256	256
DATABASE TIER				
RDS writer	A: db.r7g.large · B: db.r7g.xlarge single-AZ, 14-day PITR	1-yr Reserved, partial upfront (~35% off)	113	227
RDS read replicas	A: 1 × db.r7g.large · B: 2 × db.r7g.large separate AZ	1-yr Reserved	113	226
RDS Proxy	A: 2 vCPU · B: 4 vCPU	\$0.015 per vCPU-hour	22	44
RDS storage + backups	A: 100 GB · B: 300 GB, gp3	storage + snapshot overage	12	35
Database subtotal			260	532
CACHE				
ElastiCache for Valkey	A: 2 × cache.t4g.medium B: 2 × cache.r7g.large primary + replica, HA	1-yr Reserved nodes. Serverless was costed at \$191/mo for Tier A and lost	70	215
SERVERLESS EXECUTION				
Lambda — code execution	Code2Day sandbox, arm64	per <code>AWS_DEPLOYMENT_OPTIONS.md</code> \$5, scaled to DAU	31	78
Lambda — ASR worker	A: 79.2k jobs · B: 184.8k jobs ~10 s each at 512 MB	\$0.0000133/GB-s beyond free tier	5	12
SQS	speech job queue	first 1M requests free, then \$0.40/M	0	1
Serverless subtotal			36	91
EDGE AND STORAGE				
CloudFront	A: ~594 GB · B: ~1,386 GB egress	1 TB/mo perpetual free tier, then ~\$0.109/GB (India)	3	42
S3 — SPA, resumes, media	static assets and documents	Standard	5	10
S3 — audio recordings	A: ~250 GB · B: ~580 GB effective	90-day → Glacier IR, 12-month delete	8	18
Edge and storage subtotal			16	70

Component	Configuration	Rate basis	Tier A USD	Tier B USD
NETWORK AND OPERATIONS				
Public IPv4	~2.5 addresses average	\$3.60 per address-month. No NAT Gateway	10	10
CloudWatch	7-day log retention, core alarms	ingestion + custom metrics	20	35
Route 53, ECR, AWS Backup	hosted zone, images, vault	—	15	20
Network and operations subtotal			45	65
INFRASTRUCTURE SUBTOTAL			683	1,229
AI AND INFERENCE				
DeepSeek V3.2 — Bedrock Mantle	A: 95.1M in / 48.6M out B: 221.9M in / 113.4M out	\$0.62/M in, \$1.85/M out, ap-south-1 serverless	149	347
Groq Whisper large-v3-turbo	A: 1,320 hr · B: 3,080 hr	\$0.04 per audio-hour	53	123
AI overflow allowance	throttle and retry months	~7% contingency	15	30
AI SUBTOTAL			217	500
TOTAL PER MONTH (USD)			900	1,729
TOTAL PER MONTH (INR)			76,500	1,46,965
PER REGISTERED STUDENT (INR)			25.50	14.70
PER DAILY ACTIVE USER (INR)			25.50	21.00

5.1 Savings already applied

Lever	Tier A	Tier B
8-hour scheduled scaling + Savings Plan on baseline only (\$3)	\$469	\$469
1-year Reserved Instances on RDS and ElastiCache	\$156	\$318
Committee-of-Experts collapsed from 3 LLM calls to 1	\$81	\$189
Whisper turbo instead of large-v3	\$94	\$219
Single-AZ writer instead of Multi-AZ	\$113	\$227
Trimmed JSON output payloads	\$45	\$105
No NAT Gateway (ASR Lambda outside the VPC)	\$33	\$41
Keyword router replacing LLM intent classification	\$16	\$37
Graviton across fleet, cache and database	\$50	\$85
Audio lifecycle to Glacier IR with 12-month delete	\$17	\$40
Valkey instead of Redis OSS	\$12	\$36
CloudFront free tier absorbing SPA egress	\$62	\$109
Total saved against an unoptimised build	\$1,148	\$1,875
Unoptimised equivalent	\$2,048	\$3,604

At both tiers the savings exceed the bill that remains. An unoptimised build of the same capability would cost ₹1.74 lakh (Tier A) and ₹3.06 lakh (Tier B) a month — so more than half the money on this platform is saved by configuration and code decisions rather than spent on AWS. The two largest are both free: scheduled scaling against the 8-hour window, and reserved-capacity commitments.

5.2 What was traded away, and what it costs to buy back

Stated as accepted risks rather than omissions.

Accepted risk	Buy-back A	Buy-back B	Recommendation
Single-AZ writer. An AZ failure means promoting the replica — tens of minutes, not one	+\$113	+\$227	Take Multi-AZ before placement season; run single-AZ the rest of the year
Two nodes off-peak. Sized for residual traffic only; an unexpected evening surge queues until scaling catches up	+\$47	+\$47	Keep. Add a CPU-triggered scaling policy alongside the schedule — free
7-day log retention. Post-incident forensics beyond a week are gone	+\$15	+\$25	Keep, but export critical audit logs to S3 — cheap and long-lived
No filler-word detection. Whisper strips "um"/"uh" at every size, so the counter at <code>communication_agent.py:76</code> only matches "like/so/actually/basically"	+\$231	+\$539	Defer. Only worth it if filler accuracy becomes a defended feature
Public-subnet app nodes. Public IPs, protected by security groups rather than network isolation	+\$33	+\$41	Keep. Mitigate with strict SGs and IMDSv2 hop limit 1
12-month audio retention cap	+\$16	+\$36	Keep. Shorter retention is better privacy as well as cheaper

6 Attribution: Code2Day versus MentorMind

The estate is shared, so most of the bill belongs to neither product alone.

TABLE 4 — COST ATTRIBUTION BY PRODUCT

Attribution	Lines	Tier A	Tier B
MentorMind only	DeepSeek LLM, Groq Whisper, AI overflow, ASR Lambda, SQS, S3 audio	\$230 26%	\$531 31%
Code2Day only	Lambda code execution (the Judge0 replacement)	\$31 3%	\$78 4%
Shared platform	EC2 fleet, ALB, RDS writer + replicas + Proxy + storage, ElastiCache, CloudFront, S3 static, IPv4, CloudWatch, Route 53, ECR, Backup	\$639 71%	\$1,120 65%
Total		\$900	\$1,729

The shared majority is not MentorMind's to claim or avoid

Code2Day alone at 3,000 concurrent would still need the same ALB, the same fleet size, a database with read replicas, a cache tier and a CDN. So MentorMind's **marginal** cost is not \$230 or \$531 — it is those lines plus the share of fleet and database growth its traffic causes.

MentorMind is the request-heavier of the two — dashboards, roadmaps, resume analysis, interview turns and speech submissions, against Code2Day's submit-and-poll coding loop — so a defensible planning split is **55% of the shared tier to MentorMind**. On that basis:

All-in share	Tier A	Tier B
MentorMind (own + 55% shared)	\$581 · ₹49,385	\$1,147 · ₹97,495
Code2Day (own + 45% shared)	\$319 · ₹27,115	\$582 · ₹49,470

Splitting it precisely requires the per-endpoint request mix from nginx logs (\$8.1), which is unmeasured.

7 AI cost derivation

7.1 Speech-to-text

Groq `whisper-large-v3-turbo` , **\$0.04 per audio-hour**, billed per second with a 10-second minimum. All recordings exceed the minimum, so the headline rate holds.

Tier	Minutes/mo	Audio-hours	Cost	vs large-v3
Tier A	79,200	1,320	\$53	\$147 — saves \$94
Tier B	184,800	3,080	\$123	\$342 — saves \$219

Two required code changes, neither costing anything. `fast_mode` defaults to `False` (`communication_agent.py:13`), nothing passes it, and `views.py:555` hardcodes `speech_best` — so every request currently runs `large-v3` at ~\$0.111/hr instead of turbo at \$0.04/hr. Separately, add `response_format="verbose_json"` with word-level timestamps: pacing is derived from `file_size / 4000.0` (`communication_agent.py:72`), a byte-size guess, so every words-per-minute penalty is computed off noise.

7.2 Text LLM — DeepSeek V3.2 on Bedrock Mantle

\$0.62/M input, \$1.85/M output. Serverless, per-token, available at `bedrock-mantle.ap-south-1.api.aws` — inference stays in Mumbai. Mantle is AWS's OpenAI-compatible endpoint, so the existing OpenAI-shaped calls migrate with a base-URL and API-key change rather than a rewrite.

TABLE 5 — TOKEN VOLUME, TIER A (TIER B IS 2.33×)

Agent	Calls/mo	in	out	Input M	Output M
Speaking evaluation (consolidated to 1 call)	79,200	400	250	31.7	19.8
Interview Q&A and evaluation	26,400	1,500	600	39.6	15.8
Resume analysis	6,600	3,000	1,200	19.8	7.9
Roadmap generation	2,000	2,000	2,500	4.0	5.0
Load-balancer routing	0	moved to the existing keyword fallback, <code>load_balancer_agent.py:47</code>			
Tier A total	114,200			95.1	48.5
Tier B total	266,600			221.9	113.4

Output is 60% of the LLM bill at three times the input rate, because every agent returns JSON scores plus prose feedback. Shortening responses is worth three times as much as shortening prompts — the opposite of usual intuition.

Largest single cost risk: DeepSeek reasoning mode

DeepSeek V3.2 is reasoning-capable, and reasoning tokens bill as **output at \$1.85/M**. Every task here is structured scoring at `temperature=0.0` and needs no chain-of-thought. Left enabled, output tokens can rise 3–10×, taking the output line from \$90 to **\$270–900 at Tier A** and from \$210 to **\$630–2,100 at Tier B** — silently, with no error and no visible product change. **Pin it off and assert it in a test.**

Two Mantle specifics worth knowing

Retention. On the Responses API, `store` defaults to `true` and AWS retains full input and output for **30 days** — here that means student speech transcripts, resumes and interview answers. Data stays in `ap-south-1` and is encrypted, so residency holds, but 30-day retention of identifiable student work is a consent decision. Set `store: false` per request, or set the account data-retention mode to `none` .

Quotas. Mantle enforces *no* requests-per-minute quota — helpful for 114k–267k small calls a month — and cached input tokens do not count against the input-token quota. But DeepSeek has **no published TPM quota** on this endpoint (only Claude Opus 4.7 does); throughput is governed by internal service capacity, and Provisioned Throughput is not available on Mantle. Both tiers here need only ~12,000–28,000 input TPM sustained, which is modest, so this is a low risk at this scale — unlike at 10,000 concurrent.

7.3 Engineering prerequisites

The estimates above assume this work is done. Items 1 and 5 are hard gates.

#	Change	Why it gates the estimate
1	Async speech and resume pipeline — presigned S3 upload → SQS → Lambda → poll or WebSocket	MentorMind runs <code>gunicorn --workers 3</code> ; Code2Day runs 12 sync workers and already "collapses at ~20 concurrent submits" (profile \$1017). Synchronous AI calls at 3,000 concurrent starve logins platform-wide
2	Point the OpenAI SDK at Mantle — new base URL, Bedrock API key, IAM <code>bedrock-mantle:CreateInference</code>	Verify <code>response_format</code> with one test call first; it should be native
3	Collapse the Committee-of-Experts 3 calls → 1 (<code>communication_agent.py:125</code> , <code>chat_practice:297</code>)	Worth \$81/mo at A and \$189 at B, and cuts speech latency 3×
4	Drop <code>sentence-transformers</code> / <code>torch</code>	<code>torch==2.10.0+cpu</code> ships x86_64 wheels only; every Graviton saving in this report depends on removing it. Used in one function that already has a Jaccard fallback
5	Read routing — <code>DATABASE_ROUTERS</code> plus <code>.using()</code>	Hard gate. None exists today, so the \$113/mo and \$226/mo of read replicas do nothing until it is written
6	Materialise campus rank into Redis or a summary table	Currently three distinct-count joins in Python per dashboard load, on the writer, at up to 900 req/s
7	Set <code>CONN_MAX_AGE</code> and add RDS Proxy	A fresh connection per request exhausts the database before the instance is saturated
8	<code>FileBasedCache</code> → <code>ElastiCache</code>; <code>FileSystemStorage</code> → <code>S3</code>	Progress tracking returns wrong data across containers; audio and resumes are lost on redeploy
9	Fix <code>CORS_ALLOW_ALL_ORIGINS = True</code> (<code>main/settings.py:154</code>)	Overrides the allowlist above it, with credentials enabled. Must precede public exposure
10	Make <code>_fallback_transcript()</code> fail loudly (<code>communication_agent.py:377</code>)	Returns a hardcoded sentence that is then scored as real speech. Throttling makes this common at scale
11	Rate-limit <code>/api/communication/speaking/</code> per user	An unthrottled endpoint that bills Groq per upload makes the AI bill attacker-controlled
12	RAG → <code>pgvector</code> with an index (<code>core/rag/vector_store.py:47</code>)	Pulls every chunk into Python per query, on the writer

Also still required from `AWS_DEPLOYMENT_OPTIONS.md` Phase 0: the unauthenticated `DROP DATABASE` endpoint, unauthenticated code execution, and the forgeable password-reset token. **Nothing here is safe to expose publicly until those are closed.**

8 Verification and sensitivity

8.1 Measure these three things first

```
-- 1. Daily active users
SELECT count(*) FROM student_profiles WHERE last_login_on = CURRENT_DATE;

-- 2. Peak concurrency and the real shape of the usage window
awk '{print $4}' code2day.access.log | uniq -c | sort -rn | head -20

-- 3. Real token counts - instrument generate_completion() for one week
```

The third is the most valuable. Every LLM figure rests on estimated prompt and response sizes, and the LLM is 17–20% of the bill.

8.2 Sensitivity

If this assumption is wrong	Tier A effect	Tier B effect
Requests per concurrent user is 0.4/s, not 0.2/s	+\$310/mo	+\$420/mo
Usage window is 12 hours, not 8	+\$41/mo	+\$41/mo
Usage window is 24/7 (no scheduled scaling possible)	+\$47/mo	+\$47/mo
Voice adoption is 100% of DAU, not 40%	+\$300/mo	+\$700/mo
Token counts per call are 2× the estimate	+\$149/mo	+\$347/mo
DeepSeek reasoning mode left enabled	+\$180 to \$810/mo	+\$420 to \$1,890/mo
ap-south-1 carries a 20% uplift on AWS rates	+\$137/mo	+\$246/mo
Read routing never implemented	Replicas become dead spend; writer must scale vertically instead	

Note how little the usage window costs to get wrong in the safe direction. Even at full 24/7 operation the fleet adds only \$47/month, because the cheapest 24/7 option is simply to put all four nodes on the Savings Plan (\$259 against the chosen \$212). The expensive error is under-estimating requests per concurrent user, which drives the fleet and the database together.

8.3 Unverified inputs

Item	Status and action
DAU, peak concurrency, usage window shape	UNKNOWN Profile \$1530. Run \$8.1
<code>m7g.2xlarge</code> ap-south-1 rate	VERIFIED \$0.2333/hr
RDS, ElastiCache, ALB ap-south-1 rates	BASELINE RATES + RI DISCOUNT Confirm in the AWS Pricing Calculator; the companion document warns aggregators disagree on Mumbai
DeepSeek V3.2 price in ap-south-1	US RATE \$0.62/\$1.85 published for US East/West. Confirm in the Bedrock console
Reasoning-mode default on Mantle	UNCONFIRMED Determines the largest cost swing here. Verify, then pin
<code>response_format</code> on Mantle	INFERRED Mantle serves the OpenAI Chat Completions API. One test call decides whether item 2 in \$7.3 is trivial
40 requests/second per vCPU	ESTIMATED Load-test with k6 or Locust from inside the Region, per profile \$2271
Peak ACU requirement (for the Aurora comparison)	ESTIMATED 8–16 Only matters if revisiting \$4.2
Audio bitrate ~1 MB/min	ASSUMED Measure real <code>SpeakingSession.audio_file</code> sizes
CloudFront egress volume	ESTIMATED Depends on SPA bundle size and cache hit ratio. Tier A sits inside the free tier, so error here is cheap
<code>torch</code> on aarch64	STATED AS FAILING Confirm with a <code>pip install</code> on a Graviton instance

Two operational notes carried forward

Open a Paid-plan AWS account. Since 15 July 2025 the Free plan auto-closes after six months or when credits run out (`AWS_DEPLOYMENT_OPTIONS.md` §8 #1) — a data-loss event waiting to happen for a college.

Set AWS Budgets and a Groq spend cap on day one, with a budget action on the execution path. Neither AWS nor Groq offers a true hard ceiling, and at these token volumes a runaway loop is expensive within hours.

9 Sources

AWS documentation. Responses API — bedrock-mantle endpoint (ap-south-1 availability, authentication, `store` retention) · Quotas for the bedrock-mantle endpoint (input/output TPM, no RPM enforcement, no Provisioned Throughput) · DeepSeek V3.2 model card · Bedrock pricing · Lambda pricing · RDS for PostgreSQL pricing · EC2 on-demand pricing · ElastiCache Serverless.

Rate references. `m7g.2xlarge` ap-south-1 — \$0.2333/hr · DeepSeek V3.2 on Bedrock — \$0.62/\$1.85 per 1M · Groq pricing 2026 — Whisper turbo \$0.04/audio-hour · Aurora Serverless v2 — \$0.12/ACU-hr, scale to zero, when it saves money · ElastiCache pricing — Valkey vs Redis OSS, serverless vs node · RDS db.r7g rates · ElastiCache node rates · Deepgram Nova-3 — \$0.0043/min.

Repository documents. `AWS_DEPLOYMENT_OPTIONS.md` §§1, 5, 6, 7, 8, 9 · `PROJECT_DEPLOYMENT_PROFILE.md` §§215, 286, 362, 1017, 1530, 2271 · `MENTORMIND_MODULE_COST_BREAKDOWN.md` §§2, 4, 5, 7, 9, 10.

Code2Day + MentorMind: Cost-Minimised Deployment — combined platform cost study, 3,000 peak concurrent users at two population sizes, 8-hour daily usage window. Prepared 9 September 2026 for ap-south-1 at ₹85/USD. **Re-verify all AWS figures in the AWS Pricing Calculator before commitment.** Companion to `AWS_DEPLOYMENT_OPTIONS.md` , `PROJECT_DEPLOYMENT_PROFILE.md` and `MENTORMIND_MODULE_COST_BREAKDOWN.md` .